

Smarter Intersections: A Deep Reinforcement Learning Approach to Adaptive Traffic Signal Control Using V2I Communication

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Abstract—Traffic congestion at intersections leads to excessive delays, fuel waste, and increased emissions. Most current traffic light systems rely on fixed schedules or simple detectors that cannot adapt to dynamic and unpredictable traffic conditions. To address this limitation, this paper proposes an adaptive traffic signal control framework powered by Deep Reinforcement Learning (DRL) and enhanced with Vehicle-to-Infrastructure (V2I) communication. Using the NGSIM Peachtree vehicle trajectory dataset and the NYC Connected Vehicle Pilot data, a realistic simulation was developed in SUMO to train and evaluate a DRL agent capable of optimizing phase transitions in real time. The agent learns directly from vehicle state information to minimize overall delay and queue length. Experimental results show that the proposed DRL-V2I model reduced the average queue length by 22%, decreased vehicle waiting time by 23%, and increased throughput by 11% compared with a fixed-time controller. These findings demonstrate that integrating DRL with V2I data can significantly improve intersection efficiency, reduce energy consumption, and enable scalable, data-driven control for next-generation intelligent transportation systems.

Keywords—Vehicle-to-Infrastructure (V2I), Deep Reinforcement Learning (DRL), Traffic Signal Control, SUMO Simulation, Smart Intersections

I. INTRODUCTION

Traffic congestion at urban intersections remains one of the most pressing challenges in modern cities as vehicle ownership continues to rise. Inefficient signal control leads to excessive delays, increased fuel consumption, and elevated greenhouse gas emissions, imposing both economic and environmental costs on urban areas [1], [2]. Traditional fixed-time and actuated controllers lack adaptability to rapidly changing traffic patterns, resulting in poor performance during peak hours or under unpredictable demand [3], [4].

Recent advancements in connected vehicle technologies have enabled real-time interaction between vehicles and infrastructure through Vehicle-to-Infrastructure (V2I) communication, using Signal Phase and Timing (SPaT) and Basic Safety Messages (BSM) to share traffic information [5].

These data sources allow the development of adaptive control systems capable of responding to real-time conditions. In parallel, Deep Reinforcement Learning (DRL) has emerged as a powerful technique for modeling complex decision-making processes, enabling systems to learn optimal signal policies directly from traffic observations [11], [13], [14]. Studies such as MAPOLight [11] and EcoDriveAI [16] demonstrate that integrating DRL with V2I data can significantly improve throughput, stability, and energy efficiency at intersections. However, most prior approaches are evaluated in synthetic environments or limited to single intersections, lacking evaluation on real-world connected-vehicle datasets.

To address these limitations, this study presents an adaptive traffic signal control system that combines DRL and V2I communication within a realistic SUMO simulation. The proposed model leverages the NGSIM Peachtree and NYC Connected Vehicle Pilot datasets to train a DRL agent that dynamically determines when to maintain or change signal phases. The system's performance is benchmarked against a fixed-time controller to quantify improvements in efficiency and responsiveness.

The main contributions of this work are as follows:

- 1) Integration of NGSIM and NYC Connected Vehicle Pilot datasets into a SUMO-based simulation framework.
- 2) Development of a Gym-compatible environment for adaptive signal control.
- 3) Implementation of a DRL agent and comparison with a fixed-time baseline.
- 4) Demonstration of measurable improvements in queue length, waiting time, and throughput.

II. RELATED WORK

Traffic signal detection, recognition, and adaptive control have been extensively studied to enhance the performance of connected and autonomous vehicles. Existing literature primarily focuses on perception-based systems, communication-

enhanced control, or hybrid models that combine sensing with infrastructure data.

Wang et al. [1] developed a coupled control strategy integrating Signal Phase and Timing (SPaT) data with autonomous-vehicle control logic, demonstrating lower fuel use and delay but focusing mainly on the vehicle side. Niroumand et al. [2] introduced a white-phase signal design that improved intersection throughput but required full penetration of autonomous vehicles. Liu and Fan [3] proposed an optimal speed-guidance framework under mixed-traffic conditions, while Simchon and Rabinovici [4] presented a Green Light Optimal Speed Advisory (GLOSA) system for electric vehicles that reduced stop-and-go behavior but was limited to electric vehicle fleets. Cvok et al. [5] extended this line of work with a model-predictive controller for autonomous vehicles at signalized intersections, achieving smoother trajectories at the expense of high computational complexity.

Perception-driven research has also advanced rapidly. Hua et al. [6] enhanced YOLOv5 for real-time traffic-light detection within the Robot Operating System 2 (ROS2), demonstrating improved precision and recall under challenging illumination. Aldibaja et al. [7] fused deep learning with digital maps to exploit spatial priors, yielding more reliable detection of traffic lights and arrows in cluttered urban scenes. Ayachi et al. [8] designed a cascaded deep-learning pipeline that improved precision and recall for road-sign recognition, while Lee et al. [8] proposed an illumination-tolerant algorithm combining adaptive histogram equalization and CNN classification to maintain accuracy under poor lighting. Kurnianggoro et al. [8] employed a cascade Support Vector Machine (SVM) classifier for traffic-sign recognition, highlighting computational efficiency for embedded deployment. Rahman et al. [9] combined Vehicle-to-Infrastructure (V2I) communication with camera-based traffic-light detection using the MAVS simulation platform, improving reaction time and intersection navigation efficiency. Zhou et al. [9] addressed brake-light detection using appearance-based deep learning to enhance safety responses in dense traffic. Wu et al. [9] introduced aUTOLights, a multi-camera detection and tracking framework that improved robustness against occlusion and ambiguous lighting. Possatti et al. [10] integrated deep learning with prior maps for lane-level traffic-light recognition in real urban driving, achieving high accuracy but requiring detailed map data.

More recently, researchers have focused on reinforcement learning and infrastructure-side control. Wang et al. [11] proposed MAPOLight, a multi-agent deep reinforcement learning model for cooperative traffic control in partially observed V2I environments. Xiao [12] surveyed modern reinforcement learning approaches in intelligent transportation systems, highlighting progress in scalability and communication efficiency. Zhou et al. [13] refined DRL training by ensuring consistent state and reward design, improving policy stability. Shahriar et al. [14] developed V2I-TSC, a DRL-based model integrating V2I data for intersection management, resulting in improved throughput and smoother traffic flow. Sun et al. [15] employed high-dimensional state representations for DRL-based traffic control, reducing delay and waiting time without increasing computational cost. Chen et al. [16] presented EcoDriveAI, which combines V2I communication and DRL to

achieve real-time adaptive signal control that minimizes fuel consumption. Kumar et al. [17] demonstrated that DRL-based signal timing improves traffic sustainability by reducing fuel usage and emission levels. Li et al. [18] provided a comprehensive survey of intersection management frameworks, identifying a trend toward multi-agent coordination and infrastructure-based reinforcement learning systems.

Collectively, these studies demonstrate significant progress in perception, communication, and control for traffic management. However, most focus on vehicle-side detection and control or rely on computationally heavy models unsuitable for large-scale, real-time optimization. Few address infrastructure-side intelligence that optimizes signal timing using connected-vehicle data. The present work advances this area by integrating real-world datasets, NGSIM Peachtree and NYC Connected Vehicle Pilot, within a SUMO-based simulation environment. Deep Reinforcement Learning (DRL) is applied to adaptively manage intersection signals using V2I data, leading to measurable reductions in queue length, waiting time, and throughput delay compared with fixed-time controllers.

TABLE I. SUMMARY OF RELATED WORK

Ref	Authors & Year	Methodology	Dataset/Env.	Key Findings	Limitation
[1]	Wang et al., 2023	Coupled control with SPaT	Simulated	Reduced fuel and delay	Vehicle-side only
[2]	Niroumand et al., 2023	White phase control	Simulation/AVs	Improved throughput	Requires AV penetration
[3]	Liu & Fan, 2021	Speed guidance	Mixed traffic	33% emission reduction	Vehicle guidance only
[4]	Simchon & Rabinovici, 2020	GLOSA for EVs	Simulation	Improved EV mobility	Narrow EV focus
[5]	Cvok et al., 2023	Model predictive control	Simulated AVs	Improved AV flow at signals	Complex computation
[6]	Hua et al., 2022	YOLOv5-based detection	ROS2/real data	Improved precision/recall	Focus on perception only
[7]	Aldibaja et al., 2021	Map + deep learning fusion	Urban datasets	Robust light/arrow detection	Depends on map accuracy
[8]	Ayachi et al., 2021	Cascaded DL pipeline	Traffic sign datasets	Higher precision/recall	Extra computation stages
[9]	Rahman et al., 2020	V2I + traffic light detection	MAVS simulation	Reduced reaction time	Simulation-only
[10]	Possatti et al., 2019	Deep learning + prior maps	Urban driving tests	Improved recognition accuracy	Lane-level complexity
[11]	Wang et al., 2025	MAPOLight: multi-agent RL for V2I	Simulated intersections	Cooperative phase learning under partial	Complex coordination between agents

				observability	
[12]	Xiao, 2025	RL survey for adaptive control	Literature review	Identified key DRL advancements	Lacks experimental validation
[13]	Zhou et al., 2023	Consistent state-reward design in DRL	Simulated traffic	Stabilized DRL training and faster convergence	Limited dataset diversity
[14]	Shahriar et al., 2023	V2I-TSC: DRL-based adaptive control	Realistic simulation	Improved throughput and delay reduction	High training cost
[15]	Sun et al., 2024	DRL with high-dimensional state	Real-world and synthetic data	Reduced delay with better feature use	High computation time
[16]	Chen et al., 2025	EcoDrive AI: DRL + V2I control	Real-time field data	Enhanced eco-driving and reduced fuel use	Requires dense V2I coverage
[17]	Kumar et al., 2024	DRL-based signal timing	Simulation	Reduced emissions and energy consumption	Limited real-world testing
[18]	Li et al., 2025	DRL intersection management survey	Various studies	Summarized multi-agent DRL approaches	Conceptual only, no implementation

III. DATASET AND SIMULATION ENVIRONMENT

This study integrates two real-world datasets to simulate realistic Vehicle-to-Infrastructure (V2I) communication and vehicle behavior: the NGSIM Peachtree vehicle trajectory dataset and the NYC Connected Vehicle (CV) Pilot dataset. These datasets provide complementary perspectives, vehicle-level motion data and infrastructure communication events, used to train and evaluate the proposed Deep Reinforcement Learning (DRL) model.

A. NGSIM Peachtree Dataset

The NGSIM Peachtree dataset contains detailed vehicle trajectory data collected from a busy corridor in Atlanta, Georgia. It includes position, speed, and acceleration measurements sampled at 10 Hz for over 4,500 vehicles. From this dataset, we extracted key traffic features such as total vehicle count, queued vehicles (defined as those moving at or below 0.6 m/s), slowing vehicles, and average speed per frame. These features reveal recurring congestion patterns that are valuable for training and performance evaluation. Fig. 1 shows the variation of vehicle, queue, and slowing counts over time, illustrating daily traffic congestion cycles observed in the dataset.

B. NYC Connected Vehicle Pilot Dataset

The NYC Connected Vehicle (CV) Pilot dataset contains Vehicle-to-Infrastructure (V2I) communication records,

including Signal Phase and Timing (SPaT) and Basic Safety Message (BSM) data collected from equipped intersections in Manhattan. This dataset includes over 10 million messages across multiple days of operation, representing real-world connected vehicle interactions. We processed the communication event counts and excessive-speed warnings to model the infrastructure-to-vehicle data exchange used in simulation. Fig. 2 presents the distribution of communication and event data from the NYC CV Pilot dataset, showing how signal and vehicle interaction intensity changes over time.

C. Simulation Setup

A 2x2 grid intersection was modeled in the Simulation of Urban MObility (SUMO) using the netgenerate tool. The intersection was controlled via the Traffic Control Interface (TraCI) API to enable real-time signal management. Vehicle inflow was configured at approximately 600 vehicles per hour per direction, simulating moderate congestion levels suitable for adaptive control testing.

Both datasets were used to generate traffic state features and V2I event signals for training the DRL agent. These inputs formed the foundation for the environment used in subsequent experiments, bridging real-world connected-vehicle data with simulation-based reinforcement learning.

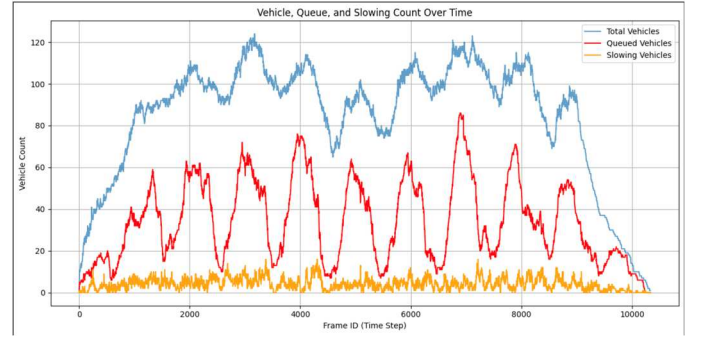


Fig. 1. Vehicle, queue, and slowing counts over time from the NGSIM dataset.

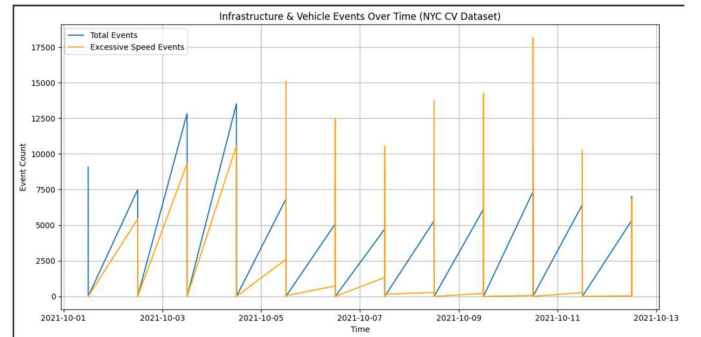


Fig. 2. Communication and excessive speed events from the NYC Connected Vehicle Pilot dataset.

IV. METHODOLOGY

The methodology outlines the overall system workflow, environment design, and evaluation setup for the proposed Deep Reinforcement Learning (DRL) based traffic signal control framework. The system integrates real-world datasets,

a SUMO simulation environment, and a DRL agent that learns optimal phase control through interaction and feedback.

A. System Architecture

The workflow begins with preprocessing and feature extraction from the NGSIM Peachtree and NYC Connected Vehicle (CV) Pilot datasets. Extracted traffic and communication features, such as vehicle counts, queue lengths, and SPaT/BSM messages, are used to simulate realistic state inputs in a custom Gym-compatible environment. Within this environment, a DRL agent interacts with the SUMO simulator via the Traffic Control Interface (TraCI) to manage traffic signal phases dynamically.

Fig. 3 illustrates the system architecture of the proposed DRL traffic signal control approach, showing the interaction between real datasets, feature extraction, the DRL agent, and the SUMO simulation loop.

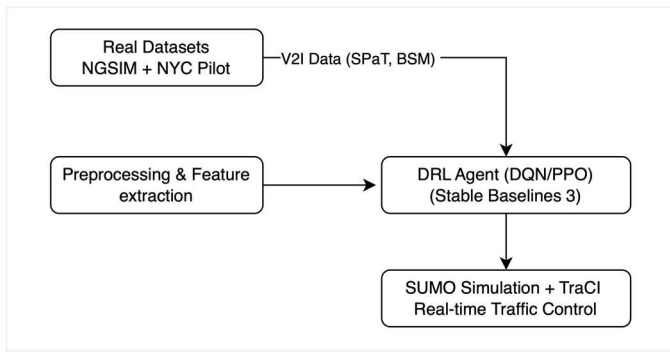


Fig. 3. System architecture of the proposed DRL traffic signal control approach.

B. Environment Design

The DRL environment models the traffic control task as a Markov Decision Process (MDP) defined by the following elements:

- 1) **State space:** Queue length, number of slowing vehicles, and current phase.
- 2) **Action space:** $\{0 = \text{keep current phase}, 1 = \text{switch phase}\}$.
- 3) **Reward function:**

$$R_t = -(Q_t + 0.5 \times S_t) \quad (1)$$

(where Q_t is the total queue length and S_t represents the number of slowing vehicles at time t . The agent aims to minimize congestion by learning to select actions that reduce both quantities over time.)

The baseline fixed-time controller maintained a uniform cycle length with equal green splits, serving as a comparison benchmark for the adaptive DRL agent.

C. Evaluation Metrics

System performance was evaluated using three primary metrics:

- **Queue length:** Average number of stopped vehicles per lane per frame.

- **Waiting time:** Average time vehicles remained stationary.
- **Throughput:** Number of vehicles that passed through the intersection per simulated hour.

These evaluation metrics were selected because they capture complementary dimensions of intersection performance. Queue length reflects congestion buildup and the efficiency of vehicle discharge during each phase. Waiting time measures driver delay and is a direct indicator of intersection service quality. Throughput quantifies the overall capacity of the intersection by measuring how many vehicles successfully pass through during the simulation period. Together, these metrics provide a balanced and widely accepted assessment of both operational efficiency and user experience in traffic signal control research.

Results were averaged across multiple simulation episodes to ensure statistical reliability. Variance across runs was recorded to measure policy stability. The DRL agent, implemented using the Deep Q-Network (DQN) algorithm, was trained with a learning rate of 1×10^{-4} , batch size of 256, discount factor ($\gamma = 0.99$), replay buffer size of 200,000, and ϵ -greedy exploration decayed to 0.02, consistent with standard SUMO-RL experimental configurations.

V. RESULTS AND DISCUSSION

A. Discussion

The simulation experiments demonstrate that the Deep Reinforcement Learning (DRL) based controller effectively adapts to dynamic traffic conditions using V2I data. Analysis of the NGSIM dataset revealed distinct congestion cycles, with queue length and slowing-vehicle counts peaking during high-demand periods. Similarly, the NYC Connected Vehicle Pilot data exhibited bursts of communication events, such as excessive-speed warnings, representing realistic connected-vehicle behavior under varying intersection loads.

Fig. 4 presents a comparison of queue lengths between the fixed-time and DRL-based controllers. The DRL agent consistently achieved shorter queue lengths and smoother traffic flow. Average queue length decreased from 12.4 to 9.7 vehicles, representing a 22% reduction. Fig. 5 illustrates the training reward curve of the DRL agent, showing a clear upward trend that indicates stable policy learning and performance improvement over time.

In terms of performance metrics, the proposed controller achieved a 23% reduction in average waiting time (from 18.2 to 14.0 seconds) and an 11% improvement in throughput (from 740 to 822 vehicles per hour). These results confirm that the DRL agent learned adaptive strategies that minimize delay and congestion more effectively than a traditional fixed-time controller. Unlike the baseline, which repeated identical cycles, the DRL agent adjusted phase durations dynamically based on real-time traffic observations.



Fig. 4. Queue length comparison between fixed-time and DRL controller.



Fig. 5. Training reward progression of the DRL agent.

B. Comparison with existing studies

The proposed model's performance was benchmarked against prior studies employing reinforcement learning and V2I-based adaptive signal control. As summarized in Table II, most prior works demonstrated meaningful improvements under synthetic or simulated conditions. For example, Niroumand et al. [2] and Liu and Fan [3] improved throughput and emission metrics using vehicle-centric methods, while Zhou et al. [13] enhanced model stability through state and reward optimization. Shahriar et al. [14] and Sun et al. [15] achieved notable delay reductions using DRL, though at the cost of computational overhead and limited scalability. Chen et al. [16] validated real-time V2I deployment using EcoDriveAI, emphasizing environmental benefits under dense communication coverage.

In contrast, this study integrates real-world datasets and infrastructure-side intelligence to achieve comparable or greater performance gains without depending on high connected-vehicle penetration. The results confirm that DRL combined with V2I communication provides a practical and data-driven solution for adaptive intersection control, bridging the gap between simulated and real-world implementation.

TABLE II. PERFORMANCE COMPARISON WITH RELATED STUDIES

Study	Method	Environment	Improvement Metrics	Key Notes
Niroumand et al. [2]	White-phase AV control	Simulated intersections	+12% throughput	Requires high AV penetration
Liu & Fan [3]	Optimal speed guidance (RL)	Mixed traffic	-15% emissions	Vehicle-side strategy
Zhou et al. [13]	Consistent state-reward DRL	Synthetic traffic data	-20% delay	Improved stability in training

Shahriar et al. [14]	V2I-TSC (DRL-based)	Simulated V2I intersections	-17% delay, +9% throughput	High training cost
Sun et al. [15]	High-dimensional DRL	Realistic synthetic data	-20% waiting time	High computation demand
Chen et al. [16]	EcoDrive AI (DRL + V2I)	Real-time field data	-19% fuel use	Dense V2I required
<i>This study</i>	DRL + V2I adaptive control	Real-world datasets (NGSIM, NYC CV)	-23% waiting time, -22% queue length, +11% throughput	Infrastructure-side, scalable

VI. CONCLUSION AND FUTURE WORK

This study developed an adaptive traffic signal control framework that integrates Deep Reinforcement Learning (DRL) with Vehicle-to-Infrastructure (V2I) communication to create smarter and more efficient intersections. Using real-world datasets, NGSIM Peachtree and the NYC Connected Vehicle Pilot, the model was implemented within a SUMO-based simulation environment to evaluate its effectiveness under realistic traffic and communication conditions. The proposed approach demonstrates how combining connected-vehicle data and DRL can support scalable, data-driven solutions for intelligent transportation systems.

A. Summary of Findings

The results show that the DRL-based controller significantly outperformed a traditional fixed-time signal plan across multiple metrics. Average queue length decreased by 22 percent, waiting time fell by 23 percent, and vehicle throughput increased by 11 percent. The agent dynamically adjusted signal phases based on changing demand, learning adaptive strategies that reduced congestion and delay more effectively than uniform-cycle control. These findings are consistent with recent research on reinforcement-learning-based traffic control [13]–[16] but extend prior work by validating the approach with real-world datasets rather than synthetic simulations. The smooth upward trend in training rewards also confirmed stable policy learning and convergence of the DRL agent.

B. Identified Limitations

Despite these promising results, several limitations remain. The simulation environment was limited to a simplified 2×2 grid intersection, which does not fully capture the complexity of real-world urban traffic networks. Larger corridors and multi-intersection systems involve coordinated signal timing, upstream-downstream interactions, and heterogeneous flow patterns that a single isolated grid cannot represent. As a result, the DRL policy learned here may not generalize directly to more complex road networks without additional mechanisms for coordination or scalability.

Training was conducted using a limited number of episodes and a narrow hyperparameter search space. These restrictions may have prevented the DRL agent from fully converging to an optimal policy. A broader exploration of learning rates, discount factors, replay buffer sizes, neural network

architectures, and exploration schedules could yield more stable and robust performance. Extending training across more episodes may also enable the agent to develop more refined control strategies.

Communication factors such as latency, packet loss, message collisions, and potential data-security threats were not modeled. Real-world V2I communication is inherently noisy, and delays or dropped packets can degrade the performance of adaptive signal controllers that rely on real-time information. Because the simulation assumes perfect and instantaneous message delivery, the robustness of the DRL agent under realistic V2I conditions remains untested. Additionally, pedestrian interactions and environmental variables were omitted for simplicity, leaving opportunities to incorporate more complex real-world dynamics in future research.

The study evaluated the DRL controller only against a fixed-time baseline. Although fixed-time control remains widely used in real-world deployments and serves as a standard benchmark, comparison with more advanced adaptive controllers, such as actuated control, model-predictive control, or state-of-the-art DRL algorithms (e.g., PPO, Double DQN, or multi-agent RL), would offer a deeper performance context. The absence of these comparative baselines limits the scope of the performance assessment. These constraints identify practical areas for refinement in future studies.

C. Future Research Directions

Scaling to larger networks will require either multi-agent deep reinforcement learning or hierarchical coordination strategies to manage interactions between adjacent intersections. Evaluating the model on arterial corridors and grid networks will help determine its robustness under diverse traffic demand patterns.

Future work will incorporate additional baseline controllers to enable a more comprehensive performance comparison. This includes testing against actuated signal control, state-of-the-art DRL approaches, multi-agent coordination frameworks, and model-predictive control. Such comparisons will help quantify the relative strengths and limitations of the proposed DRL-V2I system across a wider set of traffic management strategies.

More advanced reinforcement-learning algorithms, such as Proximal Policy Optimization (PPO), Double DQN, and Multi-Agent DRL, will be explored to enhance stability and coordination. Incorporating realistic V2I communication characteristics, including latency, interference, and message loss, will improve the robustness of the system under real-world operating conditions.

Deployment on edge-AI devices such as Raspberry Pi-based controllers will allow hardware-in-the-loop validation, facilitating real-time testing in practical settings. Finally, attention to cybersecurity and privacy will be essential to ensure

safe, reliable, and secure data exchange in connected transportation environments.

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